Parasitology in Microbiology: Beginner's Course

Lesson 3: Protozoan Parasites

Protozoan Parasites – Study Notes

Key Concepts

1. **Life cycle complexity** – Many protozoan parasites have distinct morphological stages (cyst, trophozoite, sporozoite, merozoite, gametocyte, etc.) that occur in different hosts or environments.

2. **Transmission routes** – Ingestion of cysts/oocysts, vector borne bites, or direct contact with contaminated water/soil are the main ways humans acquire these parasites.

3. **Host specificity & pathology** – Each parasite targets specific tissues (intestinal lumen, colon, red blood cells, liver) and produces characteristic clinical signs.

4. **Diagnostic forms** – Recognizing the stage that appears in stool, blood smear, or tissue biopsy is essential for laboratory diagnosis.

5. **Control measures** – Hygiene, safe water, vector control, and prophylactic drugs interrupt the life cycles.

Summary

Protozoan parasites are single celled eukaryotes that cause a wide range of human diseases. Three classic examples—*Giardia lamblia*, *Entamoeba histolytica*, and *Plasmodium* spp.—illustrate the diversity of life cycle strategies and transmission pathways.

Giardia and *Entamoeba* are intestinal protozoa transmitted via the fecal oral route. Their infective stages are environmentally resistant cysts (or, for *Entamoeba*, a mature cyst) that survive outside the host until ingested. Once in the gut, the cysts excyst to release trophozoites, which multiply and cause diarrhea (*Giardia*) or dysentery and tissue invasion (*Entamoeba*).

Plasmodium spp., the causative agents of malaria, have a far more intricate cycle involving an Anopheles mosquito vector and a human host. Sporozoites injected with the mosquito’s saliva travel to the liver, develop into schizonts, and release merozoites that invade red blood cells. Some merozoites differentiate into sexual gametocytes, which are taken up by another mosquito, completing the cycle. Understanding each stage’s location and mode of transmission is crucial for diagnosis, treatment, and prevention.

Important Points

- **Giardia lamblia**

  - **Infective stage:** Cyst (2–4 nuclei) !’ survives chlorine, cold water.
  - **Transmission:** Ingested cysts from contaminated water, food, or fecal oral contact.
  - **Pathology:** Trophozoites attach to duodenal/jejunal mucosa !’ malabsorption, watery diarrhea.
  - **Diagnosis:** Stool O&P (cysts & trophozoites) or antigen ELISA.

- **Entamoeba histolytica**

  - **Infective stage:** Mature quadrinucleate cyst.
  - **Transmission:** Same fecal oral route; often via contaminated hands or vegetables.
  - **Pathology:** Trophozoites invade colonic mucosa !’ dysentery; may disseminate to liver (amoebic liver abscess).
  - **Diagnosis:** Stool microscopy (cysts/trophozoites) + serology or PCR for extra intestinal disease.

• **Plasmodium spp. (malaria)**

  - **Key stages:**
    - *Sporozoite* (mosquito salivary glands) !’ human skin !’ liver.
    - *Merozoite* (liver schizont rupture) !’ blood stage (ring !’ trophozoite !’ schizont).
    - *Gametocyte* (blood) !’ taken up by mosquito !’ sexual reproduction !’ oocyst !’ sporozoites.
  - **Transmission:** Bite of infected Anopheles mosquito (night biting).
  - **Species & clinical relevance:**
    - *P. falciparum* – severe malaria, cerebral involvement.
    - *P. vivax* / *P. ovale* – dormant liver hypnozoites !’ relapses.
    - *P. malariae* – quartan fever cycle.
    - *P. knowlesi* – zoonotic, daily fever spikes.
  - **Diagnosis:** Thick & thin blood smears (Giemsa stain), rapid diagnostic tests (HRP 2, pLDH).

• **General transmission control**

  - Safe drinking water (boiling, filtration, chlorination).
  - Proper hand washing and food hygiene.
  - Use of insecticide treated bed nets & indoor residual spraying for malaria.
  - Prophylactic antimalarials for travelers to endemic areas.

Examples

1. **Outbreak of giardiasis in a summer camp**
   - **Scenario:** Campers share untreated stream water; several develop foul smelling, greasy stools.
   - **Explanation:** Cysts from a single infected individual survive in the water, are ingested by others, and excyst in the small intestine, leading to malabsorption. Prompt detection of cysts in stool and treatment with metronidazole stops the outbreak.
2. **Malaria case in a returning traveler**
   - **Scenario:** A traveler returns from a 2 week stay in rural Kenya, presents 10 days later with fever spikes every 48 /h, chills, and anemia.
   - **Explanation:** Infected by *Anopheles* bite !’ sporozoites !’ liver !’ blood stage *P. falciparum* (48 hour cycle). Thick smear shows banana shaped gametocytes. Immediate therapy with artemisinin based combination treatment (ACT) is required to prevent severe disease.

Quick Review

1. **What is the infective stage of *Giardia* and how is it transmitted?**
2. **Name two clinical manifestations of *Entamoeba histolytica* infection.**
3. **List the three main morphological stages of *Plasmodium* that occur in the human host.**
4. **Why are *P. vivax* and *P. ovale* capable of causing relapses?**
5. **Give one preventive measure for each parasite discussed.**

Further Reading

- Molecular diagnostics for intestinal protozoa (PCR, LAMP).
- Antimalarial drug resistance mechanisms and current ACT regimens.
- Water treatment technologies for cyst removal (ultrafiltration, UV).
- Vector biology: Anopheles mosquito behavior and control strategies.
- Host immune responses to protozoan parasites and vaccine development.